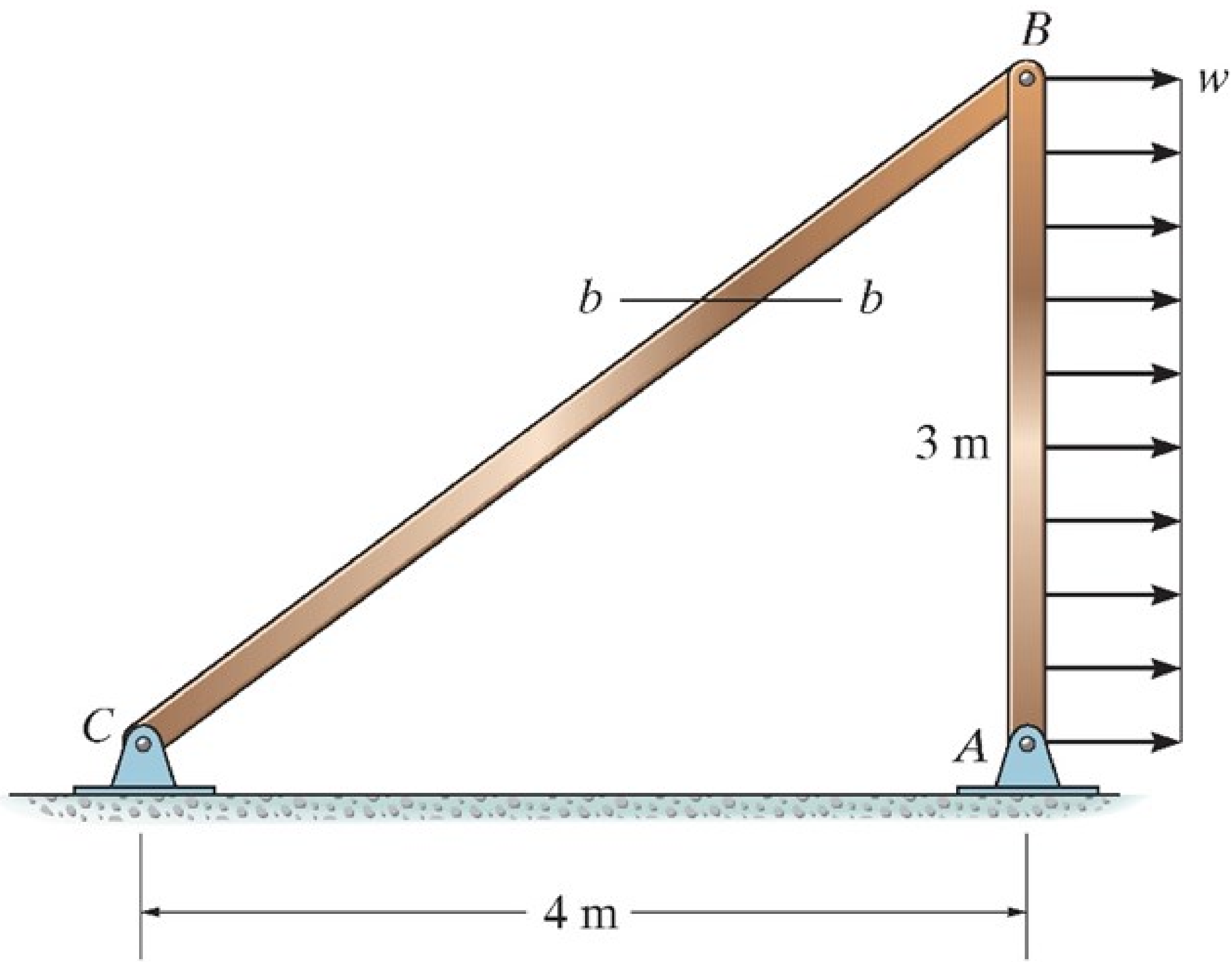
1. Determine the resultant internal force at section *b*-*b* (horizontal plane) due to the uniform loading $w = 20$ kN/m. (**15'** in total)



Solutions to Extra Exercise #1:

First, apply the free body diagram to the entire beam *AB* to get the resultant internal force in the beam *BC*:

$\sum M_A = 0 \qquad F_{BC} \times \frac{3\times 4}{5} - 3w \times 1.5 = 0$

$\rightarrow \qquad F_{BC} = 1.875w = 1.875 \times 20 = 37.5\ \text{kN}$

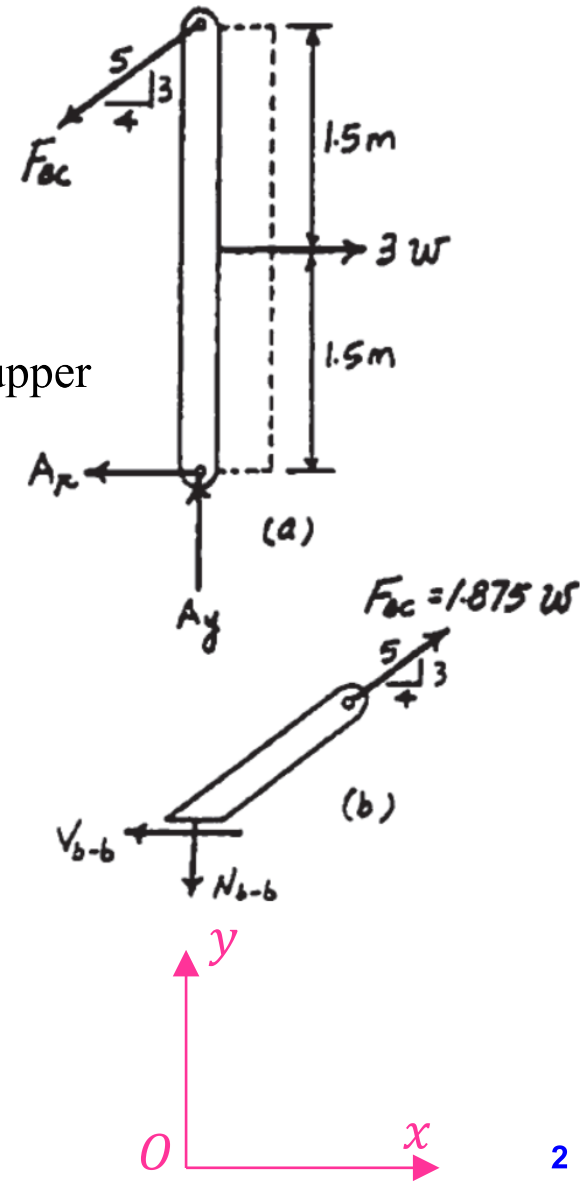
Second, section the beam *BC* along the *b-b* plane and take the upper segment to apply the free body diagram:

$\sum F_x = 0 \qquad F_{BC} \times \frac{4}{5} - V_{b-b} = 0$

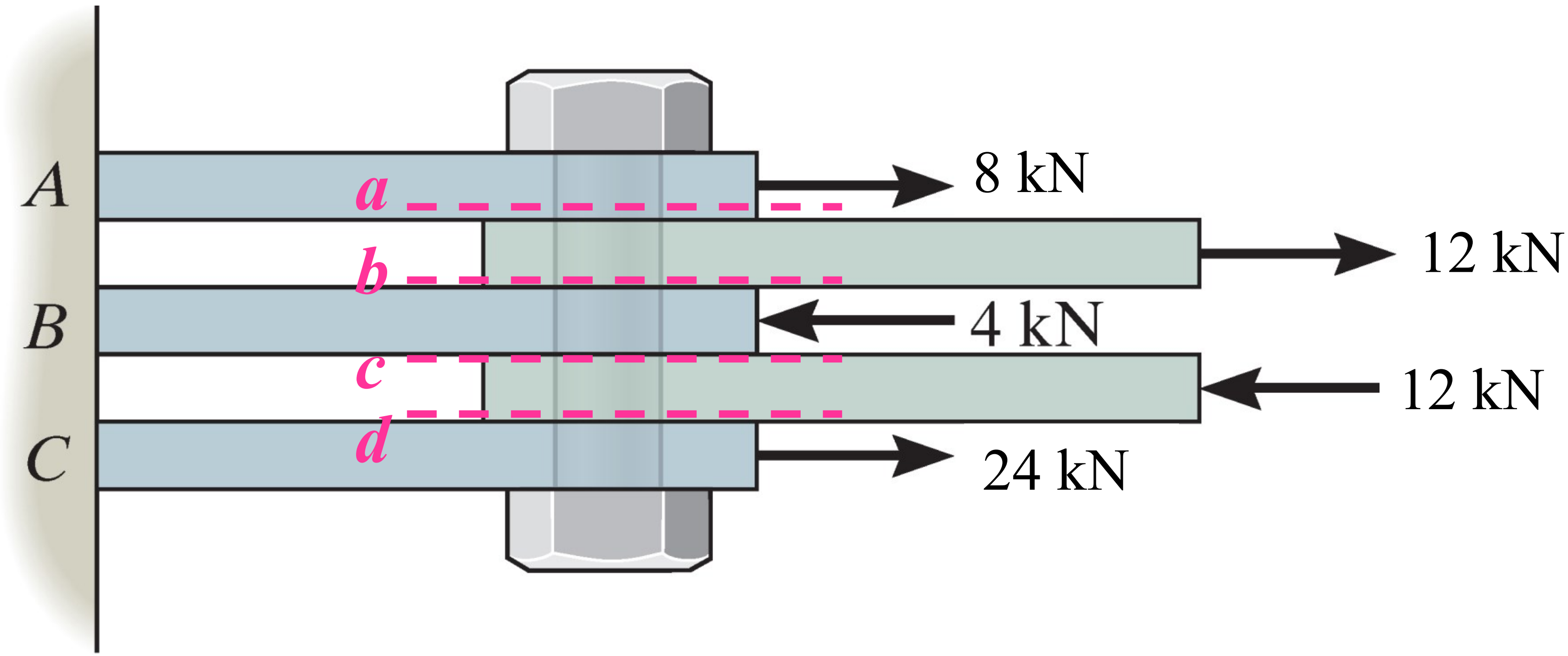
$\rightarrow \qquad V_{b-b} = 0.8F_{BC} = 30\ \text{kN}$ **(Ans.)**

$\sum F_y = 0 \qquad F_{BC} \times \frac{3}{5} - N_{b-b} = 0$

$\rightarrow \qquad N_{b-b} = 0.6F_{BC} = 22.5\ \text{kN}$ **(Ans.)**



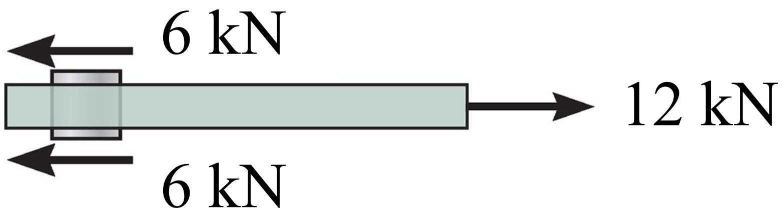
2. Determine the largest internal shear force resisted by the bolt and the reaction force at point *A*, *B*, and *C*. Include all necessary free-body diagrams. (**15'** in total)



Solutions to Extra Exercise #2:

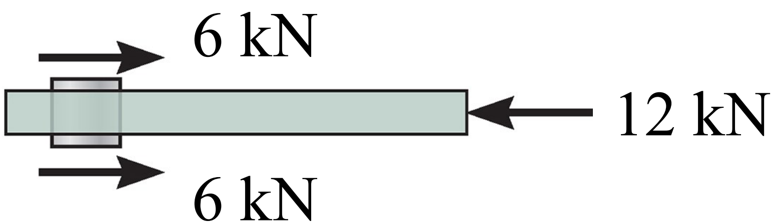
Internal shear force at plane ***a*** and ***b***:

$V = 12/2 = 6$ kN



Internal shear force at plane ***c*** and ***d***:

$V = 12/2 = 6$ kN

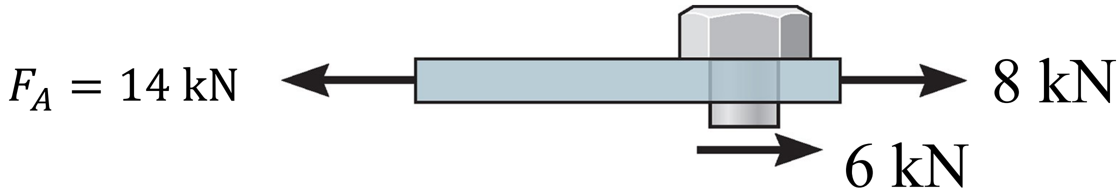


So, the largest internal shear force is 6 kN.

Solutions to Extra Exercise #2, cont'd:

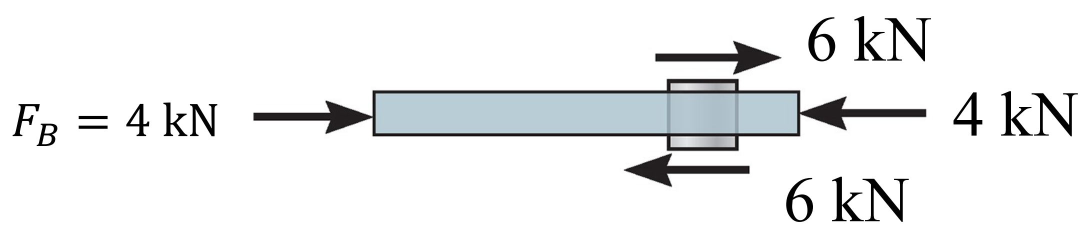
Reaction force at point ***A***:

$F_A = 8 + 6 = 14$ kN



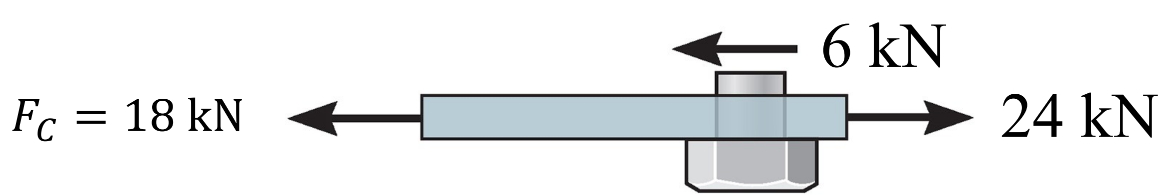
Reaction force at point ***B***:

$F_B = 4 + 6 - 6 = 4$ kN

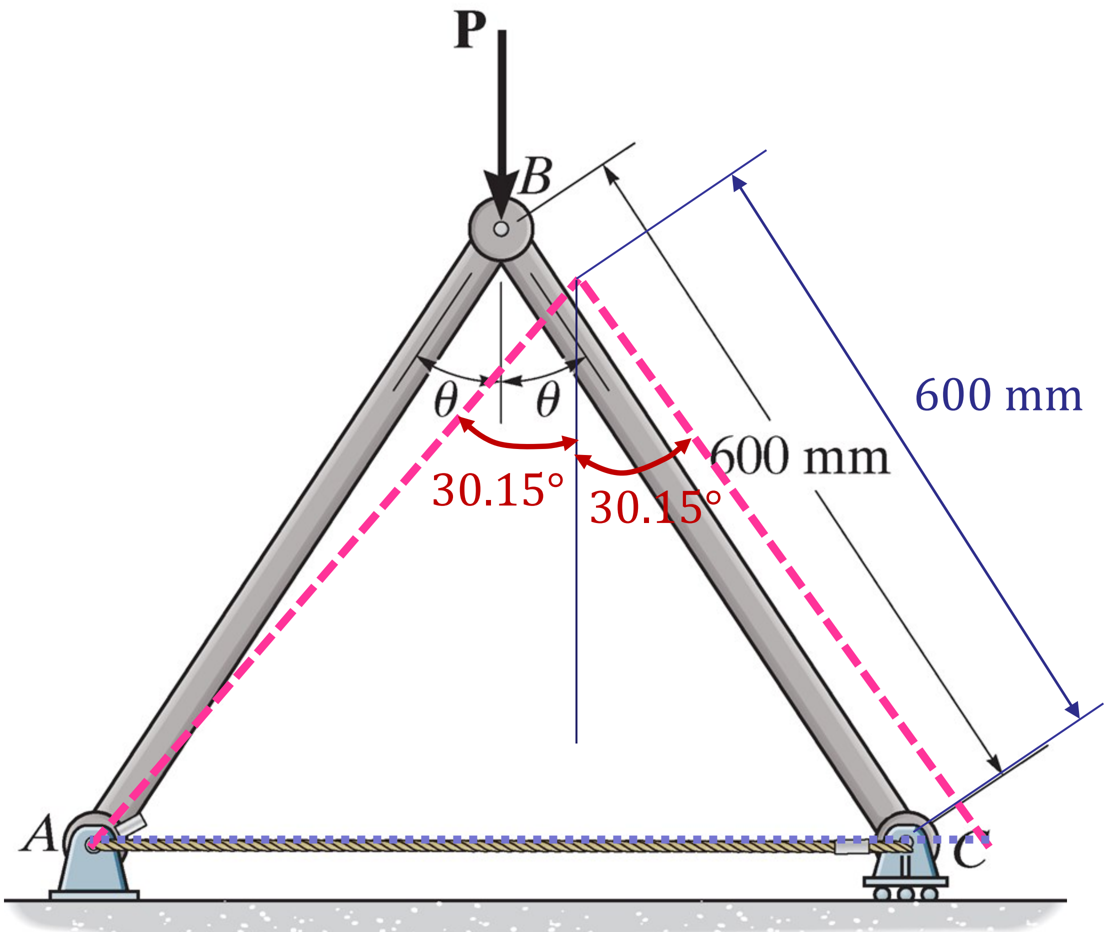


Reaction force at point ***C***:

$F_C = 24 - 6 = 18$ kN



3. The pin-connected rigid rods AB and BC are inclined at $\theta = 30°$ when they are unloaded. When the force P is applied θ becomes 30.15°. Determine the average normal strain in wire AC. (**10'** in total)



Solutions to Extra Exercise #3:

Geometry: Referring to the figure shown, the lengths of wire AC before and after stretching are

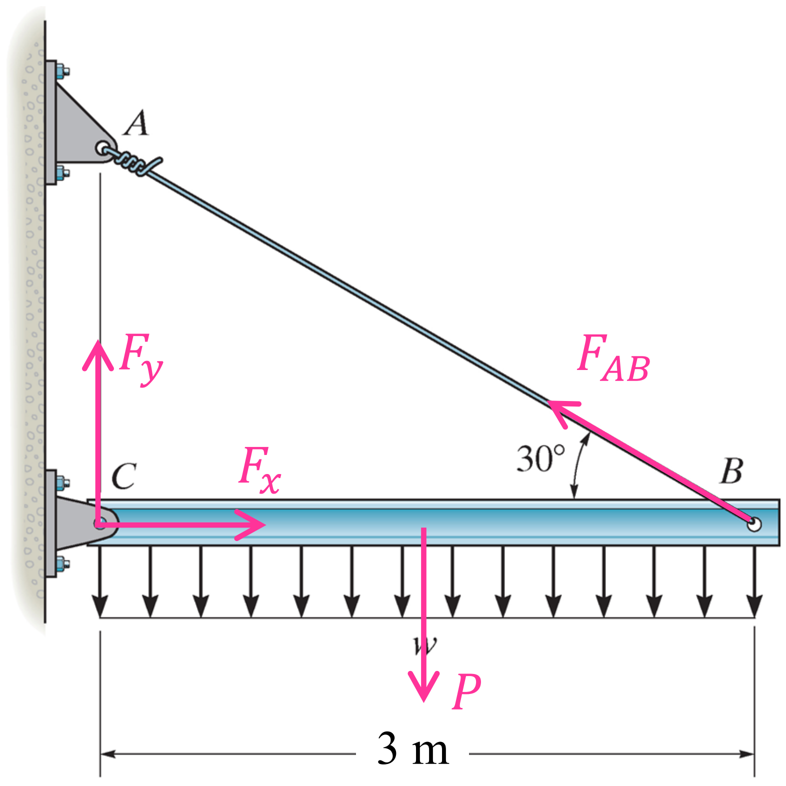
$$L_{AC} = 2 \times (600 \times \sin 30°) = 600 \text{ mm}$$

$$L_{AC'} = 2 \times (600 \times \sin 30.15°) = 602.7186 \text{ mm}$$

Average Normal Strain:

$$(\varepsilon_{\text{avg}})_{AC} = \frac{L_{AC'} - L_{AC}}{L_{AC}} = \frac{602.7186 - 600}{600} = 0.004531 \text{ mm/mm}$$ **(Ans.)**

4. The rigid beam is supported by a pin at C and a steel guy wire AB. If the wire has a diameter of 0.5 cm, determine how much it stretches when a distributed load of $w = 20\ \mathrm{kN/m}$ acts on the beam. The steel wire has Young's modulus of 200 GPa and the material remains elastic. (**10'** in total)



Solutions to Assignment #4:

In order to know the stretch (strain) in the wire, we must know the resultant internal normal force (F_{AB}) in the wire.

Applying the free body diagram to the entire beam:

$\sum M_C = 0$
→ $F_{AB} \times 3 \times \sin 30^{\circ} - 20 \times 3 \times 3/2 = 0$
→ $F_{AB} = 60\ \text{kN}$

With the resultant internal normal force (F_{AB}), we can get the normal stress in the wire:

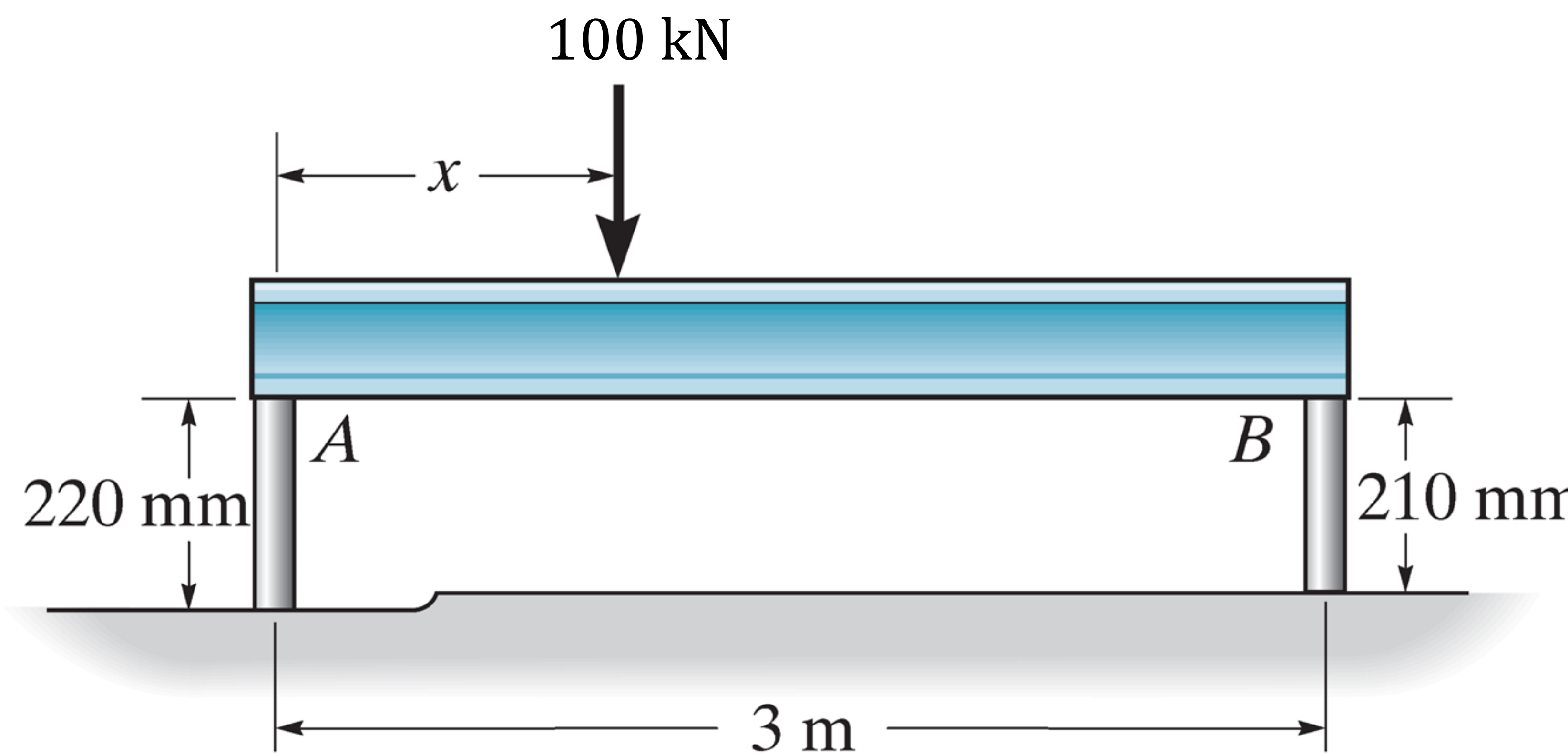
$$\sigma = \frac{F_{AB}}{A} = \frac{60000}{\frac{\pi}{4} \times 0.005^2} = 3.0558 \times 10^9\ \text{Pa}$$

According to the Hooke's law, the longitudinal normal strain is:

$$\varepsilon = \frac{\sigma}{E} = \frac{3.0558 \times 10^9}{200 \times 10^9} = 0.015279\ \text{mm/mm}$$

Then, the stretch of the wire is: $\delta = \varepsilon \cdot L_0 = 0.015279 \times \frac{3}{\cos 30^{\circ}} = 0.05293\ \text{m}$ **(Ans.)**

5. The rigid beam rests in the horizontal position on two cylinders having the unloaded lengths shown. If the two cylinders have the same diameter of 30 mm and modulus of elasticity, determine the position (i.e. placement x) of the applied load $P = 100$ kN, so that the beam still remains horizontal. What is the new diameter of cylinder A after the load is applied? $E = 73.1$ GPa, $\nu = 0.35$. **(30'** in total)



Solutions to Assignment #5:

For the rigid beam, we can draw the free-body diagram (as shown in the figure). There are two force reactions at the connection between the beam and the two bars (denoted as F_A and F_B).

The force equation of equilibrium in the vertical direction yields:

$$F_A + F_B - P = 0 \qquad (1)$$

The momentum equation of equilibrium with respect to point A gives (assuming clock-wise is positive):

$$P \cdot x - F_B \cdot L_{AB} = 0 \qquad (2)$$

Since initially (before the external force is applied) the rigid beam rests in the horizontal position, the condition that the beam still remains horizontal after the external force is applied is, the compressed displacement of the two bars (A and B) should be the same (NOTE: the strain is not the same), i.e.

$$\delta_A = \delta_B \qquad (3)$$

Using the Hooke's law and definition of strain, we can get

$$\delta_A = \frac{F_A H_A}{EA} \qquad (4a)$$

$$\delta_B = \frac{F_B H_B}{EA} \qquad (4b)$$

Solutions to Assignment #5, cont'd:

Substituting Eqs. (4a) and (4b) into Eq. (3) we have

$$\frac{F_A H_A}{EA} = \frac{F_B H_B}{EA}$$

$$F_A H_A = F_B H_B \qquad (5)$$

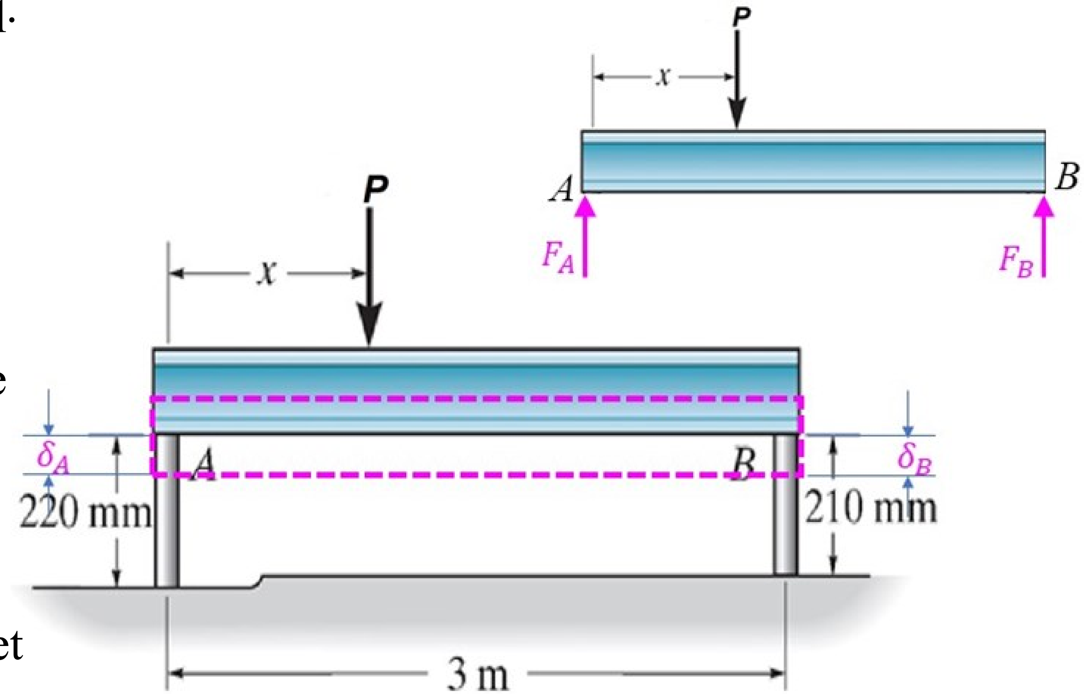
Since $H_A = 0.22$ m, $H_B = 0.21$ m, we have

$$F_A = 0.9545 F_B \qquad (6)$$

Substituting Eq. (6) into Eq. (1), we get
$F_B = 51.16$ kN

Substituting this value into Eq. (2), we finally get the position of the force (x)

$$x = \frac{F_B \cdot L_{AB}}{P} = \frac{51.16\times10^3\times3}{100\times10^3} = 1.5348 \text{ m} \qquad \textbf{(Ans.)}$$



Solutions to Assignment #5, cont'd:

The longitudinal deformation in cylinder A can be obtained by substituting Eq. (6) into (4a) (note that the stress in cylinder is compressive and thus the length is decreased):

$$\delta_A = \frac{F_A H_A}{EA} = -\frac{0.9545 \times 10^3 \times 51.16 \times 0.22}{73.1 \times 10^9 \times \frac{\pi}{4} \times 0.03^2} = -2.0791 \times 10^{-4}\ \mathrm{m}$$

The longitudinal strain in cylinder A is:

$$\varepsilon_{\mathrm{long}} = \frac{\delta_A}{L_A} = -\frac{2.0791 \times 10^{-4}}{0.22} = -9.4505 \times 10^{-4}\ \mathrm{mm/mm}$$

Using the definition of Poisson's ration, we can get the lateral strain in cylinder A:

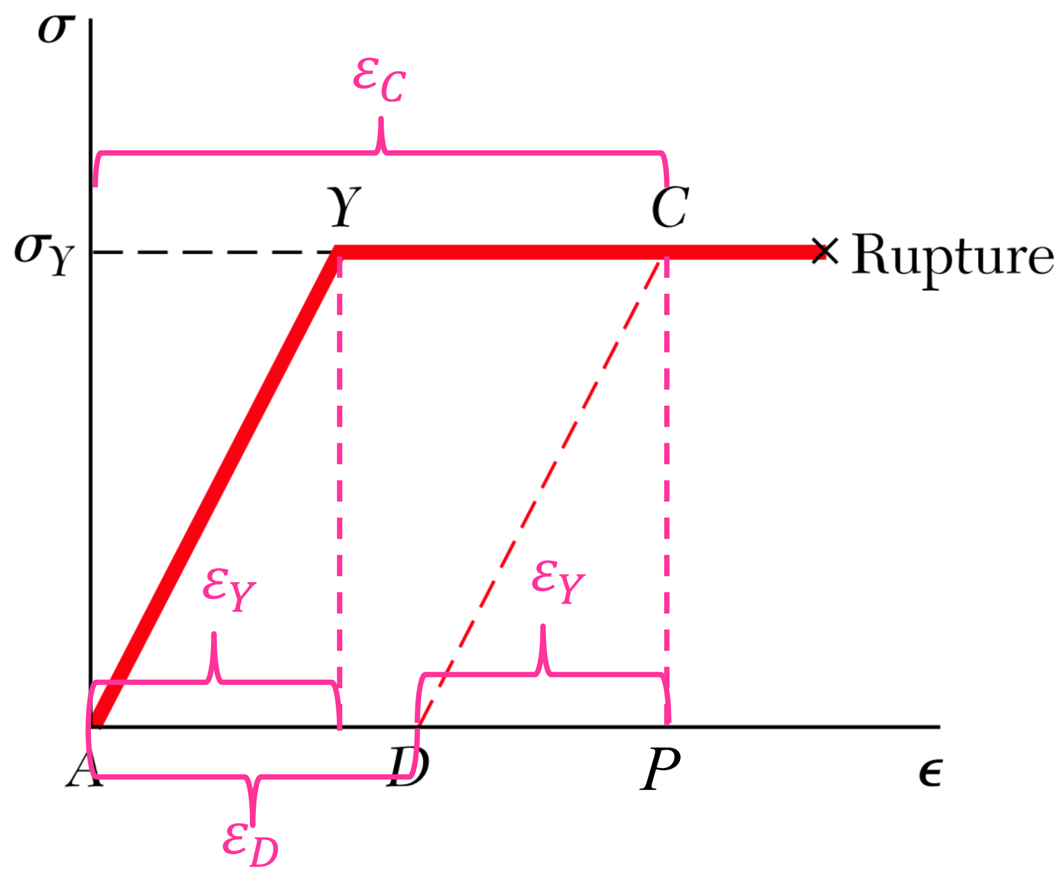
$$\varepsilon_{\mathrm{lat}} = -\nu \cdot \varepsilon_{\mathrm{long}} = -0.35 \times (-9.4505 \times 10^{-4}) = 3.3077 \times 10^{-4}\ \mathrm{mm/mm}$$

According to the definition of lateral strain, the new diameter of cylinder A is:

$$D' = (1 + \varepsilon_{\mathrm{lat}}) \cdot D_0$$

$$= (1 + 3.3077 \times 10^{-4}) \times 0.03 = 0.0300099\ \mathrm{m} = 30.0099\ \mathrm{mm}$$ **(Ans.)**

6. A rod of length $L = 500$ mm and cross-sectional area $A = 60\ \text{mm}^2$ is made of an idealized elastoplastic material having a modulus of elasticity $E = 200$ GPa in its elastic range and a yield point $\sigma_Y = 300$ MPa. The rod is subjected to an axial load until it is stretched 7 mm and the load is then removed. What is the resulting permanent set? (**20'** in total)



Solutions to Assignment #6:

Referring to the stress-strain diagram of idealized elastoplastic material , we find that the maximum strain, represented by the abscissa of point *C*, is

$$\varepsilon_C = \frac{\delta_C}{L_0} = \frac{7}{500} = 14 \times 10^{-3} \text{ mm/mm}$$

On the other hand, the yield strain, represented by the abscissa of point *Y*, is

$$\varepsilon_Y = \frac{\sigma_Y}{E} = \frac{300 \times 10^6}{200 \times 10^9} = 1.5 \times 10^{-3} \text{ mm/mm}$$

The strain after unloading is represented by the abscissa *PD* of point *D*.

According to the geometry, we note from the stress-strain diagram that

$$\varepsilon_D = AD = YC = \varepsilon_C - \varepsilon_Y = 14 \times 10^{-3} - 1.5 \times 10^{-3} = 12.5 \times 10^{-3} \text{ mm/mm}$$

The permanent set is the deformation δ_D corresponding to the strain *PD*.

$$\delta_D = \varepsilon_D L_0 = 12.5 \times 10^{-3} \times 500 = 6.25 \text{ mm}$$ **(Ans.)**